RADAR SIGNAL CLASSIFICATION

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Ministry of Education

Radar Signal Classification Using Machine Learning and Neural Networks

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Abstract

This project explores the application of machine learning and neural networks to classify radar signals. By analyzing signal features, we aim to improve accuracy and efficiency in signal identification compared to traditional methods.

Introduction

Radar signal classification is critical for defense and civilian applications. Traditional methods often struggle with complex signal environments. Our approach leverages deep learning to handle non-linear patterns effectively.

Goal

Todevelop a robust classification model capable of distinguishing between various radar signal types with high accuracy using deep learning techniques.

Dataset Description

Thedatasetconsistsofvarious radar pulse characteristics, normalized for input into the neural network architecture.

Methodology

1. Data Preprocessing.
2. Model Selection (LR, RF, KNN, DT, NN).
3. Training and Validation.
4. Performance Evaluation.

Conclusion

TheNeural Network model outperformed traditional machine learning algorithms, demonstrating superior capabilities for radar signal classification.

System Design

Implementation Tools

Python,NumPy,Matplotlib,Scikit-learn, TensorFlow/ Keras.

Results & Comparison

Model	Accuracy
Logistic Regression	0.570
Random Forest	0.748
KNN	0.759
Decision Tree	0.747
Neural Network	0.789